

Table of Contents

[DebugDefTopic](#)

[dc_displays.1.02](#)

[dc_displays.1.03](#)

[dc_displays.1.04](#)

[dc_displays.1.05](#)

[dc_displays.1.06](#)

[dc_displays.1.07](#)

[dc_displays.1.08](#)

[dc_displays.1.09](#)

[dc_displays.1.10](#)

[dc_displays.1.11](#)

[dc_displays.1.12](#)

[dc_displays.1.13](#)

[dc_displays.1.14](#)

[dc_displays.1.15](#)

[dc_displays.1.16](#)

[dc_displays.1.17](#)

[dc_displays.1.18](#)

[dc_displays.1.19](#)

[dc_displays.1.20](#)

[dc_displays.1.21](#)

[dc_displays.1.22](#)

[dc_displays.1.23](#)

[dc_displays.1.24](#)

[dc_displays.1.25](#)

[dc_displays.1.26](#)

[dc_displays.1.27](#)

[dc_displays.1.28](#)

[dc_displays.1.29](#)

[dc_displays.1.30](#)
[dc_displays.1.31](#)
[dc_displays.1.32](#)

DebugDefTopic

<	>
-----------------	-----------------

DC Debug Displays

DC Debug Displays

The DC Instrument debug displays are graphical interfaces used to assist with troubleshooting a program. A program must be validated before you can use debug displays to debug a program. The following topics are included:

- [DC30 Board Debug Display](#)
- [DC75 Board Debug Display](#)
- [DCDiffMeter Debug Display](#)
- [DCTime Debug Display](#)
- [DCVI Debug Display](#)
- [Meter Loop Tool](#)

This section describes features specific to the DC30 and DC75 DC displays. For information on using UltraVI80 displays, see [UltraVI80 Debug Displays](#).

For more information on using debug displays in IG-XL, see:

- [Using Debug Displays in TDE](#)
- [Producing Code from Debug Displays](#)

<

>

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

dc_displays.1.02



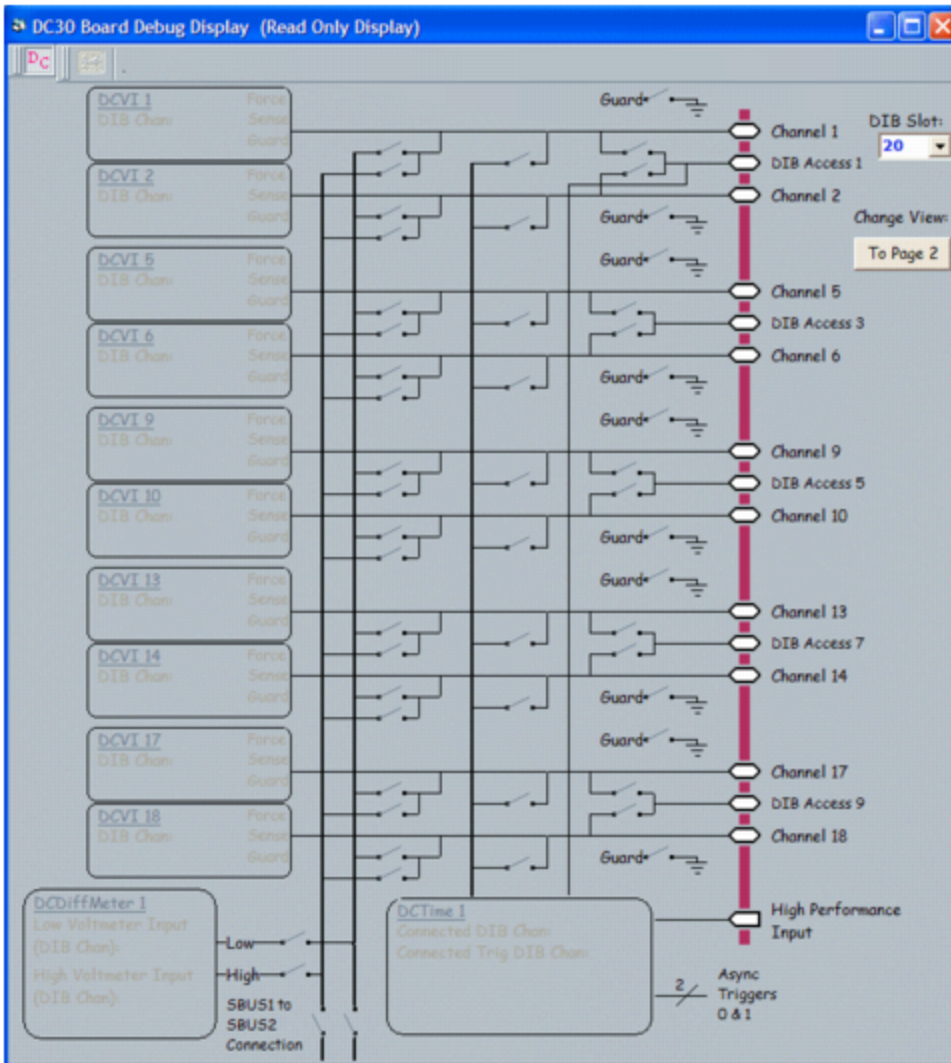
[DC Debug Displays](#) : DC30 Board Debug Display

DC30 Board Debug Display

The DC30 Board Debug Display is a read-only display showing boardwide connections. The display shows an overview of DIB access, DCVI, DC Differential Voltmeter, subbus, and DC Time Measurement Unit connections.

To view this display, on the IG-XL Tools tab. in the Launch group, select Subsystems>DC30 Board.

The figure shows an example of the display.



id

DC30 Board Debug Display

Using the Display

As a test is run, the display changes depending on the test program being executed. Channel information is highlighted on each instrument representation and relays open and close as the state of the instrument changes. The signal path is also highlighted using color-coded lines.

Click an instrument name to open the debug display for that instrument.

DIB Slot

Use this menu to select a DIB slot to display the instrument board to be debugged. Only DIB slots containing a DC Instrument board appear in this menu.

Change View

Only half the 20 channels are displayed in the display at a time. Click Change View button to display the other 10 channels.



	2015 Teradyne, Inc. All rights reserved.	
--	--	--

dc_displays.1.03



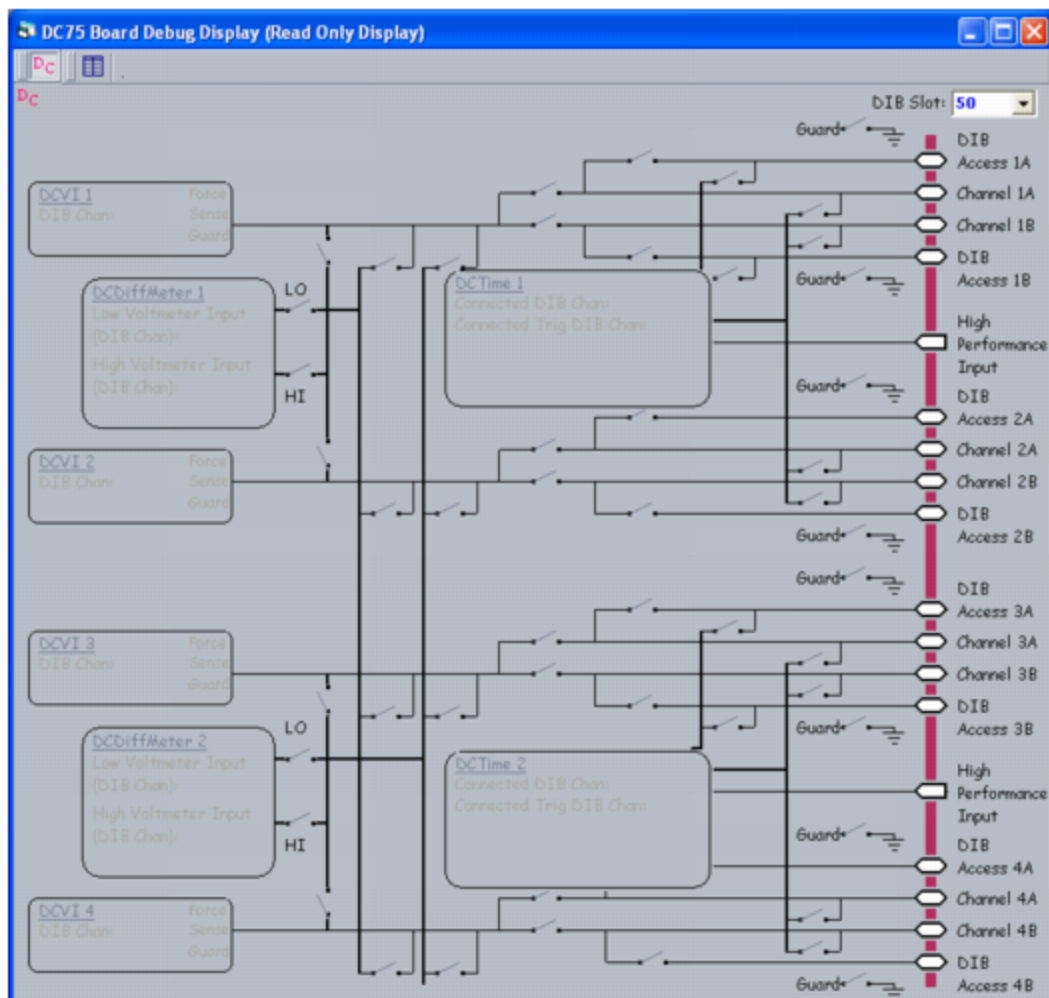
DC Debug Displays : DC75 Board Debug Display

DC75 Board Debug Display

The DC75 Board Debug Display is a read-only display showing boardwide connections. The display shows an overview of DIB access, DCVI, DC Differential Voltmeter, subbus, and DC Time Measurement Unit connections.

To view this display, on the IG-XL Tools tab. in the Launch group, select Subsystems>DC75 Board.

The figure shows an example of the display.



DC75 Board Debug Display

Using the Display

All four DCVI channels, two DCDiffMeters, and two DCTimes are shown on a single display with connections between the instruments and the DUT. Connections between merged DCVI channels are also displayed when channels are merged. Choose a DIB slot from the DIB Slot menu. As a test is run, the display changes depending on the test program being executed. Channel information is highlighted on each instrument representation and relays open and close as the state of the instrument changes. The signal path is also highlighted using color-coded lines. Click an instrument name to open the debug display for that instrument.

DIB Slot

All four channels are shown on a single display with connections to DCVI, DCTime, the DCDiffMeter, and the DUT. Choose a test head slot from the DIB Slot menu to display the instrument board to debug.

< >		
	2015 Teradyne, Inc. All rights reserved.	

dc_displays.1.04

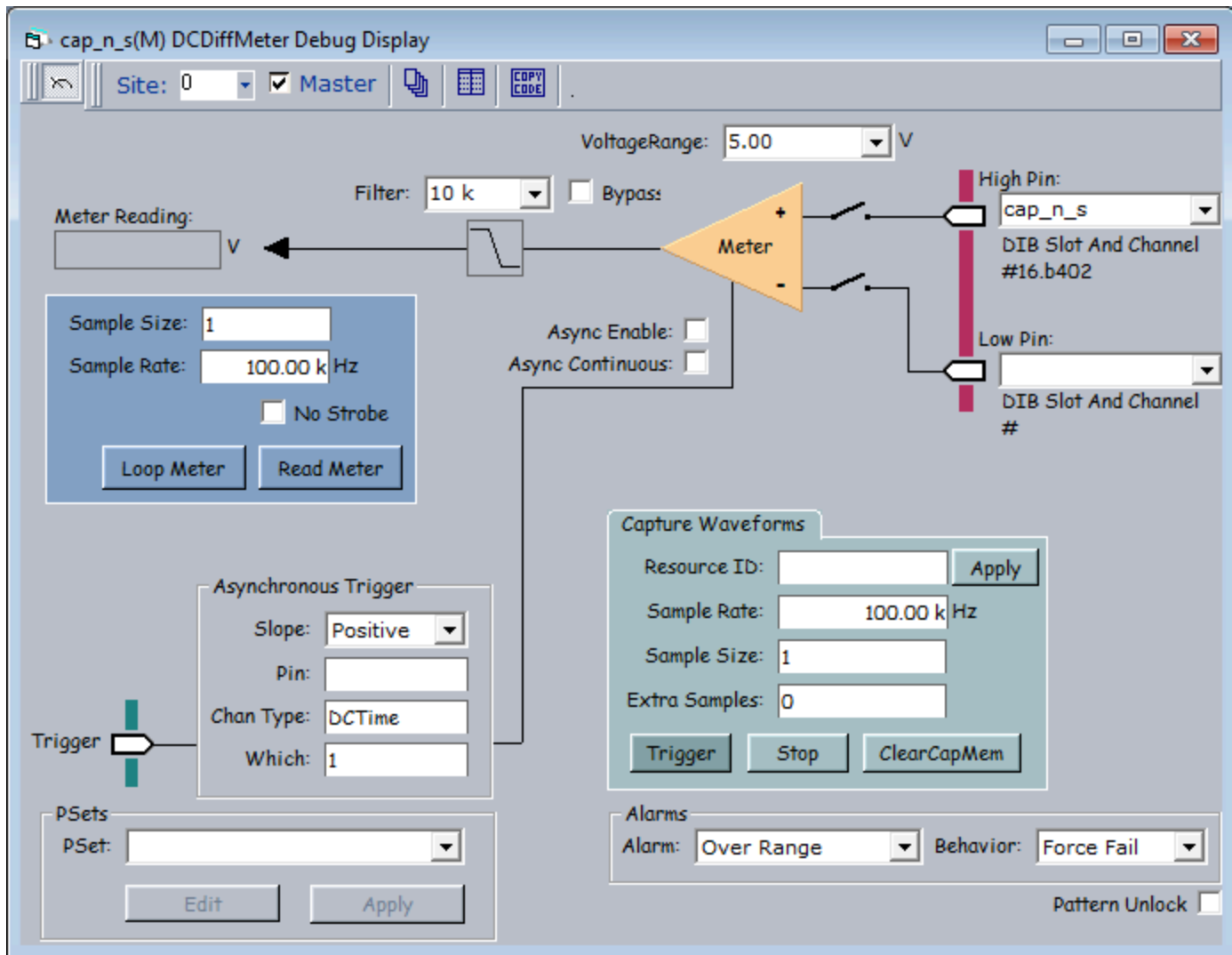


DC Debug Displays : DCDiffMeter Debug Display

DCDiffMeter Debug Display

The DCDiffMeter Debug Display is an interactive tool that shows the state of a DC Differential Voltmeter. Most parameters are interactive. Features that you can change may be limited to a certain range or selection of options provided in a menu. Noninteractive features are grayed out on the display and cannot be adjusted during debugging.

To view the DCDiffMeter Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>DCDiffMeter. The figure shows an example of the display.



DCDiffMeter Debug Display

Using the Display

To start debugging, select a pin and site to display the program parameters. Once you select a pin and site, parameters are shown in real time for that instrument.

Relay Graphics

Specifies the state of the DCDiffMeter Connection to the DUT pin. Connect or disconnect relays by clicking them.



dc_displays.1.05

<	>
-----------------	-----------------

DC Debug Displays : DCDiffMeter Debug Display : DCDiffMeter Connections

DCDiffMeter Connections

To connect or disconnect the DCDiffMeter to or from DUT pins using the debug display:

1. [Select a High Pin or Low Pin from the menu.](#)

Selecting a pin displays the DIB slot and channel that the pin is mapped to in the channel map between the two pin menus.

2. [Click relays to connect or disconnect a High or Low Pin.](#)

High Pin	Specifies the list of DCDiffMeter high pins on which the test operates.
Low Pin	Specifies the list of DCDiffMeter low pins on which the test operates.
Relay graphics	Specifies the state of the DCDiffMeter Connection to the DUT pin. Connect or disconnect relays by clicking them.

<	>
-----------------	-----------------

dc_displays.1.06

<

>

DC Debug Displays : DCDiffMeter Debug Display : Meter Setup

Meter Setup

You can change DCDiffMeter settings using the debug display. This allows you to change the program while the program is active and being stepped through. Change parameters by selecting items from a menu or clicking a control.

VoltageRange	Specifies a list of voltage ranges that can be set. A voltage range programed should be higher than the voltage level being measured.
Filter	Specifies the filter setting for the meter. See Metering Filter.
Sample Size	Specifies the number of samples to be read by the meter.
Sample Rate	Specifies the number of samples to be captured per second. This box is grayed out when No Strobe is selected.
No Strobe	Select this box to produce a reading based on the current state of the test program but using values stored in memory rather than strobing the signal. This is useful in viewing data while a pattern is running. Clear this box to cause the meter to strobe the signal based on the sample size and sample rate programmed. All results are returned to the Avg Measurement box.
Loop Meter	Opens the Meter Loop Tool and displays measurements. This button is grayed out when No Strobe is selected. See Meter Loop Tool.
Read Meter	Makes a DiffMeter reading and displays it in Meter Reading.

Pattern Unlock

Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See [PatternInstrumentUnlock](#) in the VBT Syntax Reference.

dc_displays.1.07

<

>

DC Debug Displays : DCDiffMeter Debug Display : Asynchronous Trigger

Asynchronous Trigger

Allows measurements to be made asynchronous of the test system. A change of state in the Asynchronous Trigger Pin triggers a measurement on the pin connected to the DCDiffMeter.

Asynch Enable	Select this box to have the DCDiffMeter triggered by an asynchronous trigger. If it is not selected, the asynchronous trigger signal is ignored. This box can also indicate that a trigger has occurred when Continuous mode is cleared. (See Asynch Continuous.) See Interaction with the Debug Display.
Asynch Continuous	Select this box to specify that an asynchronous trigger automatically be reenabled after the trigger event. Clear this box to require that an asynchronous trigger be manually reenabled after the trigger event.
Slope	Selects the slope of the asynchronous trigger signal. The slope can be: - Positive - Negative
Pin	Specifies the pin whose signal is used to asynchronously trigger the instrument.
Chan Type	This is the asynchronous trigger pin's channel type. The trigger channel type should match the channel type found in the channel map that

Which

corresponds with the asynchronous trigger pin.
Selects the threshold used to asynchronously trigger the instrument.
Options are:

- 1: threshold 1
- 2: threshold 2

For more information, see [Onboard Asynchronous Triggering of a DCDiffMeter](#).

<		>	
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.08

<

>

DC Debug Displays : DCDiffMeter Debug Display : Capture Waveforms

Capture Waveforms

The meter stores a number of samples based on input in memory to capture the signal. The capture memory maximum depth is 512 samples. A signal is captured based on the settings specified below.

Resource ID	Specifies a signal name to capture. Select the signal from a list of preprogrammed signals.
Apply	Loads the specified Resource ID.
Sample Rate	Specifies the number of samples to be captured per second.
Sample Size	Specifies the number of samples specified to be read by the meter.
Extra Samples	Specifies how many extra samples should be taken in the capture. For example, if the Sample Size is 3 and Extra Samples is set to 5, then the total number of samples taken is 8.
Trigger	Triggers the instrument to start the capture and opens the WaveScope.
Stop	Stops the signal capture.
ClearCapMem	Clears the capture memory.

<

>

dc_displays.1.09

<	>
-----------------	-----------------

DC Debug Displays : DCDiffMeter Debug Display : PSets

PSets

The PSet controls allow you to select an available PSet and optionally apply the settings to the DCVI hardware. You can use the following controls:

PSet	Select an available PSet. The list contains all PSets defined for the current pin.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Apply	Applies the selected PSet to the hardware for that channel.

<	>
-----------------	-----------------

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

dc_displays.1.10

<

>

DC Debug Displays : DCDiffMeter Debug Display : Alarms

Alarms

The alarm controls allow you to set behavior for one or all alarms to aid debugging. You can set the following parameters during normal debug procedures:

Alarm	Selects all alarms or a specific alarm to set an action for. Selecting an alarm only affects that alarm for that pin.
Behavior	Selects an action for the alarm or alarms selected in the Alarm menu. Alarm actions specify what should occur if the selected alarm condition exists for the pin. This menu is grayed out until you select from the Alarm menu. Options are: • Force Bin • Off • Force Fail • Default • Continue • Raise Error

< >		
	2015 Teradyne, Inc. All rights reserved.	

dc_displays.1.11

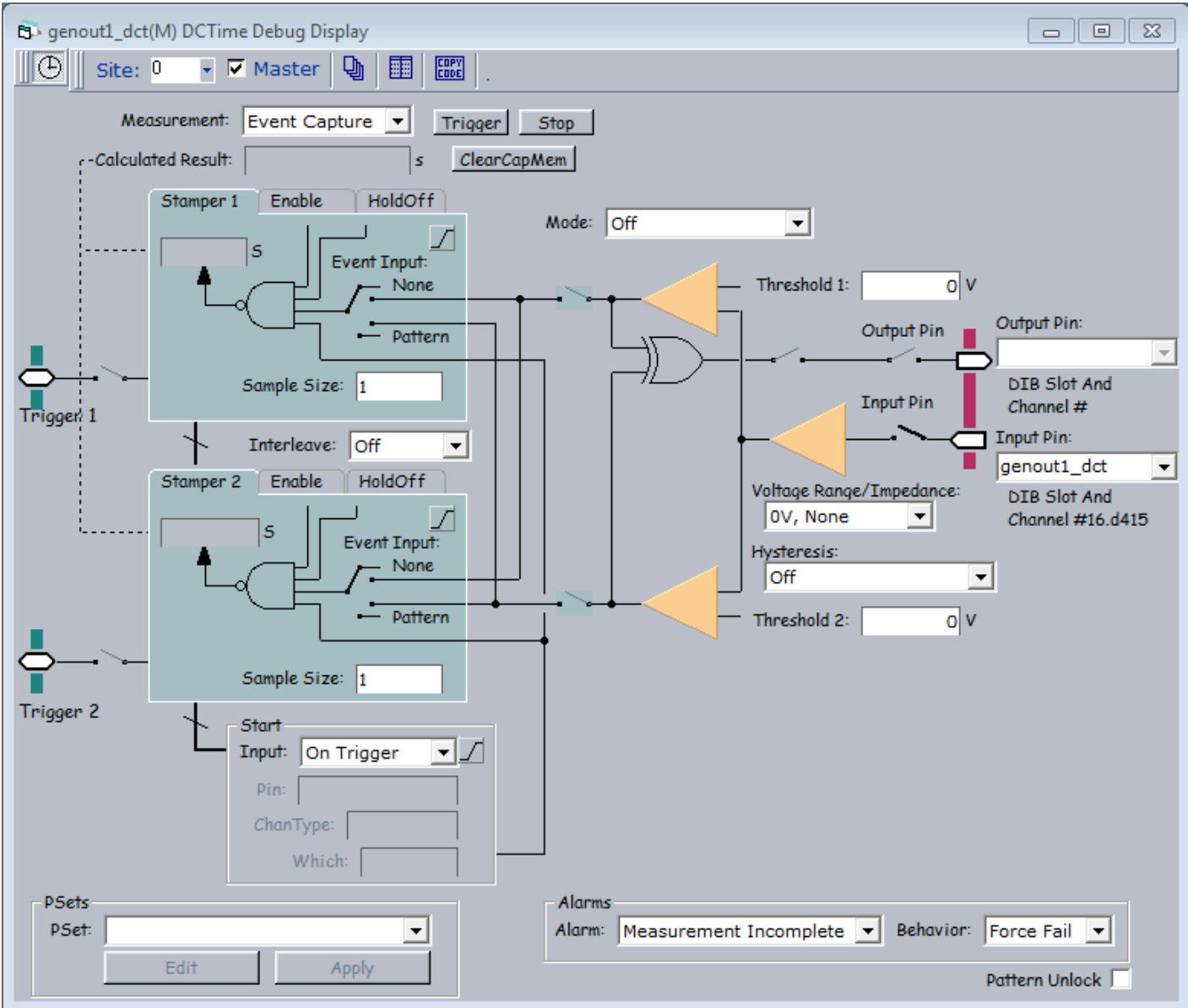


DC Debug Displays : DCTime Debug Display

DCTime Debug Display

The DCTime Debug Display is an interactive tool that shows the state of a DC Time Measurement Unit. Most parameters are interactive. Features that you can change may be limited to a certain range or selection of options provided in a menu. Noninteractive features are grayed out on the display and cannot be adjusted during debugging.

To view the DCTime Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>DCTime. The figure shows an example of the display.



DCTime Debug Display

Using the Display

To start debugging, select a pin and a site to display the program values. Select sites and pins from the menu. Once you select a pin and a site, parameters are shown for that instrument.

dc_displays.1.12

<

>

DC Debug Displays : [DCTime Debug Display](#) : DCTime Connections

DCTime Connections

Connect or disconnect the DCTime from DUT pins with the debug display using the following procedure:

1. [Select a pin from the menu.](#)

Selecting a pin displays the DIB slot and channel that the pin is mapped to below the [Pin](#) menu.

2. [Click the](#) Input Pin relay to connect or disconnect the sense lines.

Pin	Specifies the list of DCTime pins on which the test operates.
Input Pin	Specifies the state of the DCTime connection to the DUT pin. Change the relay state to connect or disconnect by clicking it.
Output Pin	Specifies the state of the DCTime connection to the DUT pin. Connect or disconnect a relay by clicking it.

<

>

dc_displays.1.13

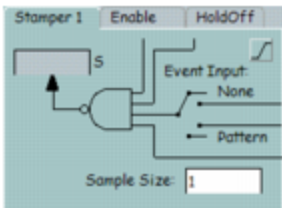
<

>

DC Debug Displays : DCTime Debug Display : Stamper Tab

Stamper Tab

Click a Stamper tab to bring its contents to the front. Choose the tab for stamper 1 or stamper 2.



DCTime Stamper Tab

You can change the following controls:

Measurement results field	Displays the measurement result for the stamper.
Sample Size	Specifies the number of samples to be read. Sample size is 1 - 512 samples.
Event Input	The event on which the time stamper is to stamp on. Click the relay to change the connection to one of the following: - None - Threshold 1 - Threshold 2 - Pattern

	See Event Signal.
Event Slope	The negative or positive slope. The icon’s picture displays either a negative or positive slope. Slope does not apply when the stamper is Off or when the input is a pattern. See Event Signal.

< >	
	2015 Teradyne, Inc. All rights reserved.

dc_displays.1.14

<

>

DC Debug Displays : DCTime Debug Display : Enable Tab

Enable Tab

Click an Enable tab to bring its contents to the front. Choose the tab for stamper 1 or stamper 2.

Stamper 1 Enable HoldOff

Input: None

Pin:

ChanType:

Which:

Mode: Gate Window

DCTime Enable Tab

You can change the following controls:

Input	Selects the event on which the stamper becomes enabled. Options are: - None - Pin - Pattern If Pin is selected, the Enable Pin field is required. Otherwise, the Enable Pin field is not required and is grayed out.
Pin (enable)	Selects a programmed pin. This is the pin that enables the time stamper. This text box is only active if the Input menu is set to Pin.
ChanType (enable pin)	The enable pin's channel type. The enable channel type should match the channel type found in the specified channel map. This text box is

	only active if the Input menu is set to Pin.
Which (enable pin)	Selects threshold 1 or threshold 2 to asynchronously enable the instrument. Options are: 1: threshold 1 2: threshold 2
Mode	This text box is only active if the Input menu is set to Pin. Selects how events are captured whenever the enable input is another pin's signal. Options are: - Gate Window: Stamper captures events as long as its enable input is at gate_high. The stamper cannot stamp if enable input is gate_low. - Trigger Each Event: Stamper must receive a trigger on its enable input before capturing the next event signal. See Enable Mode.
Slope (enable)	Sets negative or positive slope. The icon's picture displays either a negative or positive slope. Slope does not apply when the stamper is Off or when the input is a pattern.

dc_displays.1.15

<	>
----------------------	----------------------

DC Debug Displays : [DCTime Debug Display](#) : HoldOff Tab

HoldOff Tab

Click the HoldOff tab for stamper 1 or stamper 2 to bring its contents to the front.

DCTime HoldOff Tab

You can change the following controls:

Mode

Select which hold-off is used to allow time for a signal to settle before making a time measurement. Options are:

- [Precount Once](#): Counts a programmed amount of time or counts a programmed number of events before making the first time stamp on the first sample only.
- [Before Each](#): Holds off a programmed amount of time or counts a programmed number of events before making each and every stamp.
- [Off](#): No delay.

[See Hold-Off Mode.](#)

Count	Displays how many events to hold off before a signal is stamped. Hold off count waits for a certain number of events to occur before making a time measurement. See Hold-Off Count.
Time	Displays the set time before a signal is stamped. This method waits at a minimum the specified amount of time before a time measurement is made. See Hold-Off Time.

< >	
	2015 Teradyne, Inc. All rights reserved.

dc_displays.1.16

<

>

DC Debug Displays : DCTime Debug Display : Start

Start

This area of the displays controls the start signal for the DCTime.

Input	Selects the event a start responds to. Options are: - On Trigger - Pin A start is equivalent to an on/off switch for the time stamper. If the stamper has received a start signal, then the stampers are on. Otherwise, they are off. If Pin is selected, the Pin field is required. Otherwise, the Pin field is not required and is grayed out. See Start Modes.
Pin (start)	Selects from the programmed pins. This is the pin of the Start signal. This text box is only active if the Input menu is set to Pin.
ChanType (start pin)	The start pin's channel type. It should match the channel type found in the channel map that corresponds with the start pin. This box is only active if the Input menu is set to Pin.
Which (start pin)	Selects if threshold 1 or threshold 2 is used to asynchronously start the instrument. Options are:

- 1: threshold 1
- 2: threshold 2

This box is only active if the Input menu is set to Pin.

Slope (start pin) Sets negative or positive slope. The icon's picture displays either a negative or positive slope. Slope does not apply when the stamper is off or when the input is a pattern.

			
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.17

	
---	---

DC Debug Displays : DCTime Debug Display : Stamper Setup

Stamper Setup

Interleave

Options are:

- [On](#): Stamper 2 only captures events after Stamper 1 has captured to avoid misalignments of the time stamps.
- [Off](#): Interleave mode is off and stampers can stamp in any order.

[See Interleave Mode.](#)

Measurement

Displays the result of the time measurement for the stamper. This measurement is based on the selected measurement type. Each time you [click Trigger](#), you must select the correct measurement type. Options are:

- [Fall Time](#)
- [Rise Time](#)
- [Frequency](#)
- [Period](#)
- [Duty Cycle](#)

- [Pulse Width](#)
- [Event Capture](#)

[See Measurement Types.](#)

Trigger	Initiates the instrument to start a capture.
Stop	Stops the signal capture.
ClearCapMem	Clears the capture memory.
Calculated Result	Displays the result of the DCTime capture based on the Measurement.
Pattern Unlock	Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See PatternInstrumentUnlock in the VBT Syntax Reference.

<

>

2015 Teradyne, Inc. All rights reserved.

dc_displays.1.18

<

>

DC Debug Displays : DCTime Debug Display : Front End Schematic

Front End Schematic

The following are parameters in the front end schematic that can be adjusted during debug.

Threshold1/2	Specifies the value at which DCTime should either stamp or trigger.
Hysteresis	Removes noise around the threshold crossing. The gain and offset calibration constants with hysteresis turned on depend on whether you are looking for the positive or negative crossing. These constants are adjusted automatically when hysteresis type is changed. Options are: • Off • On Positive Adjust: Automatically adjusts voltage threshold for a positive slope. • On Negative Adjust: Automatically adjusts voltage threshold for a negative slope. • On: Hysteresis is on. You must take into account the shift in the voltage threshold. See Hysteresis.
Voltage Range/Impedance	Sets the voltage range and the impedance of the incoming signal so that it can be attenuated. Select a setting from the menu.

Mode

See Voltage Range and Impedance.

Determines the connection of the front end. Options are:

- Off: No connections made.
- Stamper: Connection to front end is made.
- On Board Trigger: Connection to front end and trigger is made.
- DIB Trigger: Connection to output pin is made.

<		>	
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.19

<	>
-----------------	-----------------

DC Debug Displays : DCTime Debug Display : PSets

PSets

The PSet controls allow you to select an available PSet and optionally apply the settings to the DCVI hardware. You can use the following controls:

PSet	Select an available PSet. The list contains all PSets defined for the current pin.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Apply	Applies the selected PSet to the hardware for that channel. This control is grayed out if no PSets are available.

<	>
-----------------	-----------------

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

dc_displays.1.20

<	>
--------------------------------------	--------------------------------------

DC Debug Displays : DCTime Debug Display : Alarms

Alarms

The alarm controls allow you to set behavior for one or all alarms to aid debugging.
You can set the following parameters during normal debug procedures:

Alarm	Selects all alarms or a specific alarm to set an action for. Selecting an alarm only effects that alarm for that pin. See DCTime Alarms.
-------	---

Behavior

Selects an action for the alarm or alarms selected in the Alarm menu. Alarm actions specify what should occur if the selected alarm condition exists for the pin. This menu is grayed out until you select from the Alarm menu. Options are:

- Force Bin
- Off
- Force Fail
- Default
- Continue
- Raise Error

<		>	
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.21

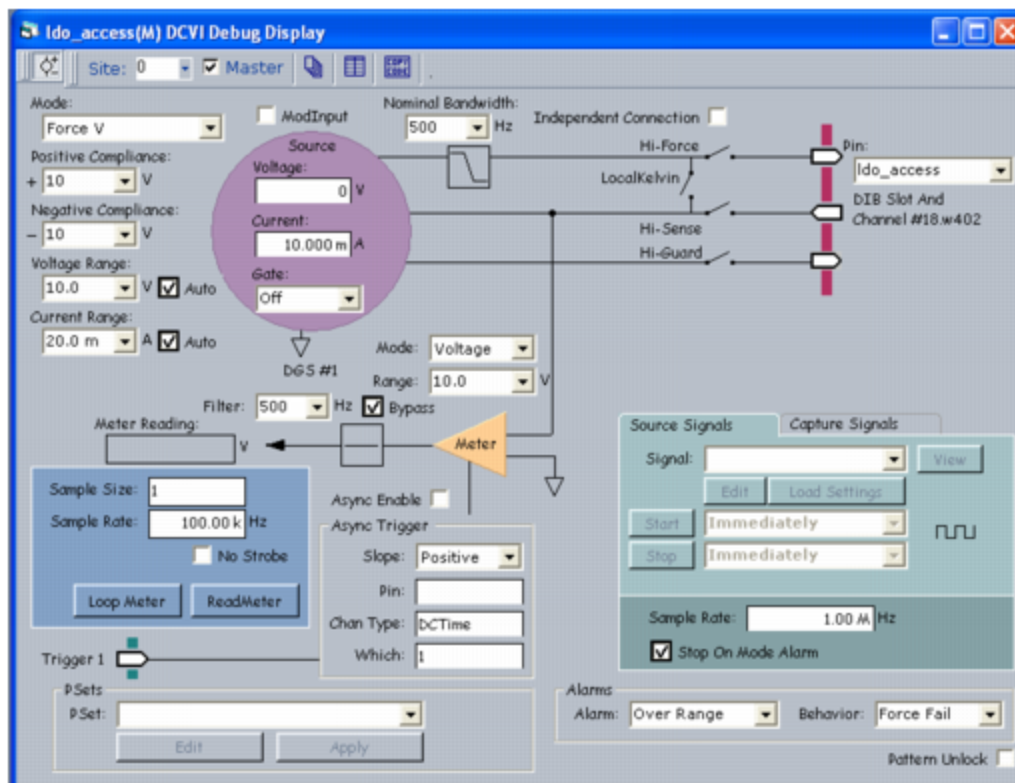


DC Debug Displays : DCVI Debug Display

DCVI Debug Display

The DCVI Debug Display is an interactive tool that shows the state of a selected DCVI channel. Features that you can change may be limited to a certain range or selection of options provided in a menu. Noninteractive features are grayed out and cannot be adjusted.

To view this display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>DCVI. The figure shows an example of the display when a DC30 or DC75 pin is selected.



DCVI Debug Display

Using the Display

To begin debugging, select a pin and site to display parameters and make changes. Select a site and a pin from the menus.

<		>	
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.22

<

>

DC Debug Displays : DCVI Debug Display : DCVI Connections

DCVI Connections

Connect or disconnect the DCVI from DUT pins with the debug display using the following procedure:

1. Select a pin from the menu.

Selecting a pin displays the DIB slot and channel that the pin is mapped to in the channel below the Pin menu.

2. Click the relays to connect or disconnect the force and sense lines.

Pin	The list of DCVI pins on which the test operates. All DCVI pins are displayed in the menu. Choosing a pin determines if the debug display is showing a DC30 or DC75 instrument.
Independent Connection	Allows you to make connections with the force and sense lines independent from each other. With the check box selected each connection relay closes independently from all others. This is not true when the check box is cleared.
Relay graphics	The state of the DCVI connection to the DUT pin. Click a relay to connect or disconnect it.

<	>	
	2015 Teradyne, Inc. All rights reserved.	

dc_displays.1.23

<

>

DC Debug Displays : DCVI Debug Display : DCVI Setup

DCVI Setup

The DCVI Schematic section displays the current programmed settings of the selected DCVI. You can change program parameters by entering a value in a text box or selecting from a menu. You can adjust the following parameters during normal debug procedures:

Positive Compliance	Select or enter a positive compliance range. Compliance ranges for the DC30 must be symmetrical. Compliance ranges for the DC75 can be asymmetrical and are programmed independently of each other. See Compliance.
Negative Compliance	Select or enter a negative compliance range. Compliance ranges for the DC30 must be symmetrical. Changing one menu automatically changes the other to match. Compliance ranges for the DC75 can be asymmetrical and are programmed independently of each other. See Compliance.
Voltage Range	Select a voltage range from the menu. Selecting Auto allows IG-XL to handle setting the voltage range automatically. Clearing Auto requires a manual entry. See Force Voltage.
Current Range	Select a current range from the menu. Selecting Auto allows IG-XL to handle setting the current range automatically. Clearing Auto requires a manual entry. See Force Current.
Voltage	The programmed voltage. Entering a valid value programs the DCVI to that value. You may need to change the range when entering a new

	voltage.
Current	The programmed current. Entering a valid value programs the DCVI to that value. It may be required to change the range when entering a new voltage.
Gate	Displays the state of the gate. Select from the following: • Off (default) • On See Gate Function.
ModInput	Selecting this box enables modulation Input. Clearing this box means that Modulation Input is not enabled and is not being used.
Mode (DCVI)	Displays the currently programmed mode, and allows you to change the mode. The DCVI must be gated off to change mode. Select a DCVI mode from the following: • Force V: See Force Voltage • Force I: See Force Current • High Impedance: See High Impedance Mode • Force V High Reg: See High Regulation Mode
Nominal Bandwidth	Displays the current programmed bandwidth setting and allows the bandwidth to be programmed to one of the selections in the menu. See Bandwidth.
Pattern Unlock	Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See PatternInstrumentUnlock in the VBT Syntax Reference.

<		>	
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.24

<

>

DC Debug Displays : DCVI Debug Display : DCVI Meter Setup

DCVI Meter Setup

Each DCVI channel has a dedicated meter. Use the meter to make measurements on DUT pins through proper relay connections and isolations. The channel meters can strobe the measurement asynchronously by communicating with the onboard trigger lines. These lines are driven by the outputs of the DCTime front-end circuits.

You can adjust the following parameters during normal debug procedures:

Mode	Specifies the mode of operation of the DCVI meter as one of the following: - Voltage - Current
Range	Specifies the metering range of the DCVI meter. This menu switches between current and voltage based on the Mode selected. See DCVI Meter.
Filter	Specifies the filter setting for the DCVI meter. Select from the menu: 500 10k See Metering Filter.

Bypass	Select this box to set the meter filter to bypass. Clear this box to use the filter setting in the Filter menu.
Sample Size	Specifies the number of samples to be read by the meter. See DCVI Meter .
Sample Rate	Specifies the number of samples to be captured per second. This box is grayed out when No Strobe is selected. See DCVI Meter .
No Strobe	Select this box to produce a reading based on the current state of the test program but use values stored in memory rather than strobing the signal. This is useful in viewing data while a pattern is running. Clear this box to cause the meter to strobe the signal based on the sample size and sample rate programmed. See DCVI Meter .
Loop Meter	Opens the Meter Loop Tool and displays measurements as long as the meter is making measurements. This button is grayed out when the No Strobe is selected. See Meter Loop Tool .
Read Meter	Makes a DCVI meter reading and displays it in Meter Reading label.

<

>

2015 Teradyne, Inc. All rights reserved.

dc_displays.1.25

<	>
-----------------	-----------------

DC Debug Displays : DCVI Debug Display : Source Signals

Source Signals

A signal can be sourced to a DCVI pin. Enabling Autoload allows IG-XL to enter a preprogrammed signal automatically and disables the Load and Unload buttons. If Autoload is cleared, you must manually load a signal or an error occurs.

Signal	The menu includes the names of the defined signals. If a default signal has been defined, the default signal’s name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Brings up the WaveScope tool and plots the sourced signal segment on the WaveScope display. This allows you to visually preview and analyze the signal that is to be sourced. When you are running the program on a test system, WaveScope displays the samples and scales them by the amplitude. When working offline, the WaveScope displays the ideal waveform.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Load Settings	Specifies If the driver computes and loads the register settings needed to create the signal named in the Signal menu.
Start	Starts the signal based on the setting specified in the menu next to the button:

- **Immediately**: The signal starts and loops automatically each time it ends.
- **Once**: The signal is sourced and stops at the end.

The square wave is a sourcing indicator that moves when the source is sourcing on the tester. This feature only works on line.

Stop Stops the signal based on the setting specified in the menu next to the button:

- **Immediately**: The signal terminates at that point.
- **At End**: The signal stops at the end of the current cycle.

Sample Rate Specifies the number of samples to be sourced per second.

Stop On Mode Alarm Select this box to terminate signal sourcing if an alarm occurs. If unselected, the program continues based on the alarm behavior setting.

< >		
	2015 Teradyne, Inc. All rights reserved.	

dc_displays.1.26

<

>

DC Debug Displays : DCVI Debug Display : Capture Signals

Capture Signals

The meter stores a number of samples based on input in memory to capture the signal. The capture memory maximum depth is 512 samples.

Signal	The menu includes the names of the defined signals. If a default signal has been defined, the default signal’s name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Brings up the WaveScope tool and plots the captured signal segment on the WaveScope display. This allows you to visually preview and analyze the signal that is to be captured. When you are running the program on a test system, WaveScope displays the samples and scales them by the amplitude. When working offline, the WaveScope displays the ideal waveform.
Load Settings	Specifies if the driver computes and loads the register settings needed to create the signal named in the Signal menu.
Trigger	Triggers the instrument to start the capture.
Stop	Stops the signal capture.
Sample Rate	Specifies the number of samples to be captured per second.
Sample Size	Specifies the number of samples specified to be read by the meter.
Extra Samples	Specifies how many extra samples to take in a capture. For example, if Sample Size is 3 and Extra Samples is set to 5, then the total number of

samples taken is 8.

ClearCapMem

Clears the capture memory.

			
		2015 Teradyne, Inc. All rights reserved.	

dc_displays.1.27

<

>

DC Debug Displays : DCVI Debug Display : Asynchronous Trigger

Asynchronous Trigger

Allows measurements to be made asynchronous of the test system. A change of state in the Asynchronous Trigger Pin triggers a measurement on the pin connected to the DCVI meter.

Asynch Enable	Select this box to have the DCVI Meter triggered by an asynchronous trigger. If it is not selected, the asynchronous trigger signal is ignored. This box can also indicate that a trigger has occurred when Continuous mode is not enabled. See Interaction with the Debug Display.
Slope	Select the slope of the asynchronous trigger signal. Options are: - Positive - Negative
Pin	Specifies the pin whose signal is used to asynchronously trigger the instrument.
Chan Type	The asynchronous trigger pin's channel type. The trigger channel type should match the channel type found in the channel map that corresponds with the asynchronous trigger pin.
Which	The threshold used to asynchronously trigger the instrument. Options are: 1: threshold 1

- [2](#): threshold 2

<		>	
		2015 Teradyne, Inc. All rights reserved.	

dc_displays.1.28

<	>
-----------------	-----------------

DC Debug Displays : DCVI Debug Display : PSets

PSets

The PSet controls allow you to select an available PSet and optionally apply the settings to the DCVI hardware. You can use the following controls:

PSet	Select an available PSet. The list contains all PSets defined for the current pin.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Apply	Applies the selected PSet to the hardware for that channel. This control is grayed out if no PSets are available.

<	>
-----------------	-----------------

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

dc_displays.1.29

<	>
-----------------	-----------------

DC Debug Displays : DCVI Debug Display : Alarms

Alarms

The alarm controls allow you to set behavior for one or all alarms to aid debugging. You can set the following parameters during normal debug procedures:

Alarm	Select all alarms or a specific alarm to set an action for. Selecting an alarm only effects that alarm for that pin. See DCVI Meter
-------	--

Behavior

Select an action for the alarm or alarms selected in the Alarm menu. Alarm actions specify what should occur if the selected alarm condition exists for the pin. This menu is grayed out until you select from the Alarm menu. Options are:

- Force Bin
- Off
- Force Fail
- Default
- Continue
- Raise Error

<		>	
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.30

<

>

DC Debug Displays : Meter Loop Tool

Meter Loop Tool

The Meter Loop Tool is a standard IG-XL tool that allows you to loop a meter without disabling the debug display from which you started looping. You can also:

- [Loop multiple channels](#)
- [Monitor all results in one place](#)
- [Start or stop all channels simultaneously](#)
- [Start WaveScope for any pin](#)

This section includes the following topics:

- [Starting Meter Loop Tool](#)
- [Using the Meter Loop Tool](#)

<

>

	2015 Teradyne, Inc. All rights reserved.	
--	--	--

dc_displays.1.31



DC Debug Displays : Meter Loop Tool : Starting Meter Loop Tool

Starting Meter Loop Tool

Start the Meter Loop Tool from the IG-XL Tools tab or from an instrument's debug display.

To start the tool from the IG-XL Tools tab, in the Launch group, select Tools>MeterLoop. When you start the tool in this manner, most controls are unavailable until you begin looping measurements on a pin.

To start the Meter Loop Tool from an instrument's debug display:

1. Open the debug display for the pin and instrument you want to meter.

Debug displays that call the Meter Loop Tool include DCVS, DCVI, and DCDiffMeter.

2. Set up the conditions in the debug display as you want to loop them.

3. Click Loop Meter.

The Meter Loop Tool opens and begins reading the meter for the pin you selected in the debug display. The Loop Meter button changes to Stop Looping.

4. To stop looping do either of the following:

- Click the Stop Looping button in the debug display.

- Click the red stop symbol in the Meter Loop Tool toolbar.

Repeat this procedure for each pin you want to monitor in the Meter Loop Tool.

<		>	
	2015 Teradyne, Inc. All rights reserved.		

dc_displays.1.32

<

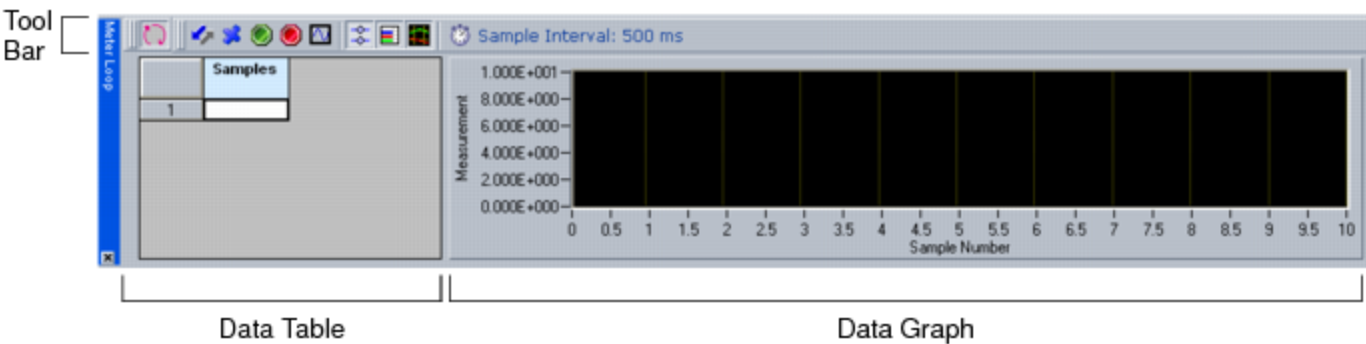
>

DC Debug Displays : Meter Loop Tool : Using the Meter Loop Tool

Using the Meter Loop Tool

The Meter Loop Tool displays measurement information in tabular and graphical formats. The display has the following sections:

Tool bar	Includes controls for customizing and managing the Meter Loop Tool. See Tool Bar Controls.
Data table	Shows the currently selected pins and, if you are looping, a history of measurements for each pin. Each pin's readings appear in a single column or row. Each column or row includes a color-coded label that identifies the source type (such as DCVS) and the channel (the channel name and the site number). You can scroll the Data Table to view earlier measurements.
Data graph	Shows looping information graphically, with sample numbers and ranges as axes. The color of each pin's line in the graph matches the color of the pin's label in the data table.



Meter Loop Tool

Tool Bar Controls

You can customize and manage the Meter Loop Tool using the following tool bar controls:

Swap Data Table Axis



Switches between displaying sample data as columns or rows. Default is to show sample data as columns.

Clear Display



Empties the columns and graphs of meter readings and resets the loop count. Does not stop or pause looping.

The Meter Loop Tool also clears the display when you add a pin to the tool.

Begin/Resume Looping



Begins (if there is no existing loop) or resumes (if you have paused a loop and have not made changes on any debug display since you paused the loop) meter readings.

Stop/Pause Looping



Stops each display from reporting measurements to the Meter Loop Tool. This is the equivalent of clicking Stop Looping on each display. This does not clear the measurement history, so you can click Begin/Resume Looping () to continue looping.

You can click Stop Looping on the channel's debug display to pause looping for that channel. If you are monitoring multiple channels, you must click Stop Looping on each channel's debug display.

If you are looping multiple channels, you can stop looping for individual channels without affecting the others. For more information, see [Removing a Source](#).

Choose Meter Plot



Selects a plot to open in WaveScope. Click this icon to display a list of available pins. Click the name of a pin to open that pin's loop waveform in WaveScope. See [About WaveScope](#).

Show/Hide Graph Points



Shows or hides points where the measurement line crosses a grid line. Default is on.

Show/Hide Data Table



Show/Hide Graph



Shows or hides the measurement table. Disabling the table makes viewing complicated measurement graphs clearer. Default is on.

Set Sample Rate



Shows or hides the measurement graph. Disabling the graph makes viewing large pin sets clearer. Default is on.

Sets the rate at which debug displays report measurements to the tool. The current rate appears to the right of this control. The default rate is 500ms.

Removing a Source

To remove a channel from the Meter Loop Tool’s display, right-click the column header of the channel you want to remove, then select Stop Looping and/or Remove.

If the selected channel is currently looping, the tool stops looping for that channel.

If the selected channel is stopped, the tool removes that channel from the display.

Copying Cell Data

You can copy data from a cell or range of cells, or an entire column. It is easier to copy data if you pause the loop.

To copy data from a cell or range of cells, select the cell or cells you want to copy and press Ctrl+C.

To copy an entire column, double-click the column header which contains the pin type and channel number, then press Ctrl+C.

< >	
	2015 Teradyne, Inc. All rights reserved.